Feasibility study

Feasibility Study

- Feasibility study can be defined as an evaluation or analysis of the potential impact of the proposed project.
- After request clarification, analyst proposes some solutions. After that for each solution it is checked whether it is practical to implement that solution. This is done through feasibility study. In this various aspects like whether it is technically or economically feasible or not. So depending upon the aspect on which feasibility is being done it can be categorized into four classes:
  - Technical
  - Economic
  - Legal
  - Operational
- The outcome of the feasibility study should be very clear. It should answer the following issues.
  - Is there an alternate way to do the job in a better way?
  - What is recommended?

1. Technical Feasibility

- Technical feasibility study checks if the software can be developed with the current equipment, existing software technology and available personnel.
- In technical feasibility the following issues are taken into consideration.
- Whether the required technology is available or not
    - Whether the required resources are available:

    - Manpower- programmers, testers & debuggers

    - Software and hardware
- Once the technical feasibility is established, it is important to consider the monetary factors also. Since it might happen that developing a particular system may be technically possible but it may require huge investments and benefits may be less. For evaluating this, economic feasibility of the proposed system is carried out.

2. Economic Feasibility

- Economic feasibility study checks if there are sufficient benefits in creating the new system to make cost acceptable.
- For any system if the expected benefits equal or exceed the expected costs, the system can be judged to be economically feasible. In economic feasibility, cost benefit analysis is done in which expected costs and benefits are evaluated. Economic analysis is used for evaluating the effectiveness of the proposed system.
- In economic feasibility, the most important is cost-benefit analysis. As the name suggests, it is an analysis of the costs to be incurred in the system and benefits derivable out of the system

3. Operational Feasibility

- Operational feasibility study checks whether the new system will be used when developed and implemented.
- Operational feasibility is mainly concerned with issues like whether the system will be used if it is developed and implemented. Whether there will be resistance from users that will effect the possible application benefits? The essential questions that help in testing the operational feasibility of a system are following.

" Does management support the project?

" Are the users not happy with current business practices? Will it reduce the time (operation) considerably? If yes, then they will welcome the change and the new system.

" Have the users been involved in the planning and development of the project? Early involvement reduces the probability of resistance towards the new system.

" Will the proposed system really benefit the organization? Does the overall response increase? Will accessibility of information be lost? Will the system effect the customers in considerable way?

4. Legal Feasibility

- Legal Feasibility study checks if there are legal hassle in developing the system.
- It includes study concerning contracts, liability, violations, and other legal traps frequently unknown to the technical staff. Whatever the system, it has to be legally approved before we implement the system.

5. Schedule feasibility

- Schedule feasibility study checks if the project deadlines are reasonable and whether constraints placed on the project schedule can be reasonable met.
- It is the study designed to determine the practicability of a system or plan.
- It is the probability of a project to be completed within its scheduled time limits, by a planned due date.
- If a project has a high probability to be completed on time, then its schedule feasibility is appraised as high.
- Schedule feasibility study includes use of the following:
  - Gantt and PERT charts.
  - Critical path method (CPM).